The Rise of Artificial Intelligence in Banking Research: A Two-Decade Bibliometric Analysis with Structural Break Detection

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Abstract

Artificial intelligence is widely said to be transforming banking, yet existing bibliometric evidence is descriptive, rests on small screened samples, and predates the generative-AI shock. We quantify the rise using 498,958 banking-domain articles (2000–2025) from Semantic Scholar, classifying 15,451 as AI-relevant via transparent keyword matching - an upper bound on true volume, since a stratified 100-record audit yields pooled precision of 55%. AI output grew 176-fold in count and from 0.5% to 13.9% of field share. Chow tests at two pre-declared breakpoints reject on all four series, confirmed by wild bootstrap. The statistic cannot by itself locate a date: on raw counts it rises monotonically across the whole admissible range, which is what convexity produces, and every candidate split from 2013 to 2020 rejects. On the curvature-free log specifications the profile has a genuine interior maximum, at 2016 on log shares and 2017 on log counts, localising a growth-regime transition to the 2016–2017 window rather than to a single year. The post-2022 generative-AI acceleration is real but weaker than the raw statistic implies: it holds on counts ($F = 105.9$), falls by an order of magnitude on logs ($F = 5.5$), vanishes within a stable-venue panel, and coincides with a 23.6% contraction in total banking publishing, so much of the field-share movement is denominator arithmetic. Its clearest expression is compositional: unclassified work fell from 50% to 12% while regulation and compliance grew six-fold. AI papers are cited at several times the rate of their non-AI peers

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within every era, and venue participation rose from 172 to 5,663 outlets even as leadership concentrated.

Keywords: Artificial intelligence, banking, bibliometrics, Semantic Scholar, structural break, research trends

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1. Introduction

Artificial intelligence has moved from the margins of banking research to its center. In 2015, machine-learning terms appeared in roughly half of one percent of banking-domain articles; by 2025 the figure was close to fourteen percent. Practitioner surveys and reviews describe this shift qualitatively [10, 13], and several bibliometric studies have charted early growth [17, 1, 7]. Yet the existing evidence has three gaps. It is largely descriptive, reporting curves without testing when - or whether - growth actually changed pace. It predates or stops at the generative-AI shock of late 2022, leaving the largest candidate discontinuity unexamined. And it works from small screened samples (typically 400-1,100 articles), which limits what can be said about an entire field.

This paper measures the rise directly. We assemble a corpus of 498,958 banking-domain research articles published between 2000 and 2025 from the Semantic Scholar open database [25], classify each as AI-relevant through transparent keyword matching on titles and abstracts, and analyze the resulting 15,451 AI papers against four research questions:

RQ1 How did the volume and field-share of AI research in banking evolve over 2000–2025?

RQ2 Does the series exhibit structural breaks aligned with the deep-learning wave (2015–2017) and the generative-AI shock (2022)?

RQ3 Which application topics drove the rise, and how did composition shift after 2022?

RQ4 Do AI papers earn a citation premium, and how concentrated is the venue landscape?

Our contributions are:

1. An evidence base two orders of magnitude larger than any screened review of this literature - 15,451 AI-classified articles drawn from a corpus of 498,958 banking-domain articles - with every number generated by a re-runnable analysis pipeline. Classification precision is audited on a stratified sample, so the counts are interpretable as upper bounds. The audit finds no significant difference in precision across eras, which is what licenses treating misclassification as noise common to all years - though a null result in a 100-record audit is weak evidence, and we treat this assumption as the least secure link in the chain.
2. Evidence that the rise is not smooth, reported with an explicit account of what the statistic can and cannot establish. Both pre-declared breaks reject, confirmed by wild bootstrap. We then show that the Chow statistic does not by itself identify a date on a convex series - every candidate split from 2013 to 2020 rejects, and on raw counts F rises monotonically across the whole admissible range - and that the curvature-free log specifications carry a genuine interior maximum, localising a growth-regime transition to the 2016–2017 window.
3. A post-2022 generative-AI acceleration that is real but substantially weaker than the raw statistic suggests: it attenuates by an order of magnitude on logs, vanishes within a long-established-venue panel, and coincides with a contraction in total banking publishing. Its footprint lies mainly in what banking researchers study rather than how much they publish.
4. Evidence of field maturation: unclassified "general" AI work fell from 50.3% to 11.7% of output across the topic windows while regulation and compliance rose six-fold, and AI papers are cited at several times the rate of their non-AI peers within every era. The medians behind that comparison are small, so we report the per-era premium as descriptive rather than as a tested effect.
5. A reproducible pipeline (cached API responses, fixed queries, unit-tested statistics) enabling exact replication.

Section 2 situates the study; Section 3 documents data and methods; Section 4 answers each question, with Section 4.5 collecting the sensitivity analyses; Section 5 interprets; Section 6 concludes.

2. Related Work

2.1. AI in Finance: Reviews and Landscapes

Comprehensive reviews establish that AI techniques now span credit risk, fraud detection, trading, customer service, and compliance in financial services. Cao [10] provides the densest technique- centered landscape, contrasting classic and modern AI for financial data, while Dixon et al. [13] systematize machine-learning methodology for finance practitioners. These works are narrative or conceptual: they describe breadth but do not measure the pace of the field's transformation.

2.2. Bibliometric Studies of AI in Finance

Four bibliometric analyses are closest to ours. Goodell et al. [17] mapped research clusters through co-citation and bibliographic coupling; Ahmed et al. [1] reviewed 348 Scopus-indexed articles; Bahoo et al. [7] screened Web of Science records down from 1,218 to 892 eligible studies, of which 110 informed their content analysis, and Pattnaik et al. [30] applied PRISMA screening to reach 1,045 included studies. The Bahoo review covers WoS-indexed work only to March 2021 - explicitly before the generative-AI shock. Adjacent mappings cover explainable AI in finance [11], fintech disruption [5], and a narrative synthesis of the banking-fintech evidence base [35]; systematic reviews of AI across banking and broader financial services [22, 27] document the same acceleration qualitatively, while expert-based frameworks position AI within banks' digital-transformation decisions [33]. Methodologically our design follows the bibliometric-analysis guidelines of Donthu et al. [14] in combining an output-level mapping with a screened analytical subset, and the science-mapping tradition more broadly [39, 4]. Three limitations recur: samples are small relative to the field (hundreds to low thousands after screening), analysis is descriptive (publication curves, co-citation maps), and observation windows end at or before 2022 - although Chen et al. [11], published in 2023, does reach into the generative-AI period. None of these works tests the growth series for statistically detectable structural breaks, which is the specific gap this paper addresses. We make no claim that no bibliometric study anywhere has applied break detection to publication data; our search was not a systematic review of the econometrics-of-science literature.

2.3. Trend and Break Methods

Our statistical toolkit follows standard practice: Mann-Kendall trend testing with Sen's slope [26, 24, 34] for monotonic growth under non-normal

series, Chow tests [12] in the structural-break tradition of Perron [31] and Zivot and Andrews [38], extended to multiple breaks by Bai and Perron [8], Holm correction [23] across the break-test family, and Gini coefficients with bootstrap intervals [15] for venue concentration. Undated breakpoints are handled with the supremum-F distribution and its wild bootstrap [3, 20], which we use in place of the Chow statistic's asymptotic null whenever the break year is not fixed in advance.

Two adjacent literatures bear on our results. Work on asymptotic growth in science systems [36] documents near-exponential growth in article counts across fields and countries, which is precisely why our log-scale re-estimation matters: on a raw scale, a break statistic fitted to an exponentially trending series partly measures curvature rather than a change of regime. And the scientometric finding that research-system expansion does not reliably coincide with rising citation impact [2] bears directly on how we read our own citation results, as we note in Section 5. Applying formal inferential break detection to bibliometric series is, to our knowledge, novel in this literature, and that novelty is a limitation as much as a contribution (Section 5).

2.4. Data Sources in Bibliometrics

Database choice shapes any bibliometric result: Scopus and Web of Science differ systematically in journal, field, and language coverage [28], Google Scholar trades curation for breadth [21], curated sources such as Scopus carry their own selection and metadata-quality trade-offs [6], and open scholarly graphs such as OpenAlex maximize coverage at the cost of noisier metadata [32]. Semantic Scholar shares the open graph's strengths and weaknesses; we therefore report corpus construction queries in full, audit classification precision on a stratified sample, and probe database-coverage drift with a stable-venue panel (Section 4.5).

3. Data and Methods

3.1. Data Source and Corpus Construction

We draw on the Semantic Scholar open database [25], accessed through its bulk-search API. The denominator corpus - all banking-domain research - is defined by the frozen query `banking | "commercial bank" | fintech | "financial institution" | lending` restricted to works published 2000–2025, returning 498,958 records after deduplication (titles, abstracts, venues, publication years, citation counts). Records lacking a year or outside the

window are excluded at ingestion; deduplication keys on Semantic Scholar paper identifiers.

The numerator - AI-relevant banking research - is every record whose title or abstract matches any of eight frozen term families, listed in Table B.7. Matching is case-insensitive substring containment. Keyword classification is transparent and exactly reproducible; its known cost is recall error for papers using unlisted terminology, which we treat as measurement noise common to both eras (Section 5). The resulting subset contains 15,451 papers (3.10% of the corpus). The eight families were chosen to cover the terminology in general use across the whole window rather than tuned to any period; Section 4.5 reports what changes when eleven further families are added.

Queries were executed once with responses cached verbatim; every table and figure regenerates from cached data without network access.

3.2. Measures

The headline measure is AI's *field share*: the fraction of banking-domain articles in year t classified as AI-relevant, $s_t = A_t/N_t$, where A_t counts AI papers and N_t all papers. Shares normalize away growth of publishing volume itself, though not entirely: they are sensitive to the level of the denominator, which turns out to matter for the 2022 result (Section 4.3). Secondary measures are yearly counts A_t, application-topic composition via keyword buckets (credit and lending risk; fraud and security; trading and markets; customer service and NLP; regulation and compliance; other), venue counts and their Gini coefficients per era [15], and median citations within era cohorts.

Eras follow candidate technology waves rather than data-fitting: an early period 2000–2010, a middle period 2011–2017 preceding mainstream deep learning [18], and the modern period 2018–2025; the break tests below evaluate - rather than assume - the corresponding turning points.

3.3. Statistical Analysis

Trend: Mann-Kendall tests [26, 24] with Sen's slope estimator [34] on yearly counts and shares, using the tie-corrected variance and a continuity correction. Breaks: Chow tests [12] comparing a pooled linear trend against segment-specific lines at each pre-declared breakpoint. A break labelled T partitions the 26 observations into a pre-break segment 2000–T and a post-break segment $T+1$–2025, so the unrestricted fit carries four parameters and F is reported on $(2, 22)$ degrees of freedom. Testing uses the pre-declared

breakpoints (2017, 2022), with Holm correction [23] across the family of four tests; this family is the only *confirmatory* inferential set in the study. Because the series is short ($n = 26$) and positively serially correlated, the two pre-declared tests are additionally reported with wild-bootstrap p-values [20] computed from 20,000 draws with Rademacher weights, generated under the null of no break using restricted residuals. Those bootstrap values, not the asymptotic F values, are what we rely on for the confirmatory family; the asymptotic ones are reported alongside for comparability. All remaining procedures - log-scale re-estimation, endogenous scans, double-break grids, stratified samples, per-bucket tests, venue-panel checks, and denominator-constant re-estimation - are *robustness and exploratory* analyses; their p-values are reported descriptively and are not corrected for multiplicity. Robustness: every break test is re-estimated on logarithmic counts and log shares, since output compounds approximately exponentially and a raw-scale break statistic therefore partly measures curvature; an endogenous supremum-F scan [3] with wild bootstrap inference [20], generating its null from restricted-residual distortions under the null of no break, probes undated breakpoints; a Bai–Perron-style grid search over breakpoint pairs [8] checks whether further breaks are supported; stratified samples excluding preprint and rapid-review venues probe composition artifacts; and a stable-venue panel (outlets with ≥ 10 banking-domain papers in 2000–2009 *and* ≥ 25 in 2018–2025) re-estimates both breaks within outlets already indexed before the deep-learning wave, probing database coverage drift - though its late-period criterion means it selects for outlets that stayed active, which we note where we use it. A denominator-constant variant recomputes field share with N_t held at its 2020 value, separating numerator growth from movement in total banking-domain output. Citation premium: Mann-Whitney U comparisons of AI versus non-AI citation counts within cohorts aged at least three years (2000–2010, 2011–2017, 2018–2021), to limit right-censoring of recent work; given medians of 1–5 these distributions are heavily tied, so we report them as descriptive rank comparisons rather than as effect-size estimates. Venue concentration: Gini coefficients with percentile-bootstrap confidence intervals (2,000 resamples, resampling venues, which are the Gini's units). Significance level $\alpha = 0.05$ throughout; bootstrap p-values from 1,000 replicates are reported at their floor, $p \leq 1/1001$.

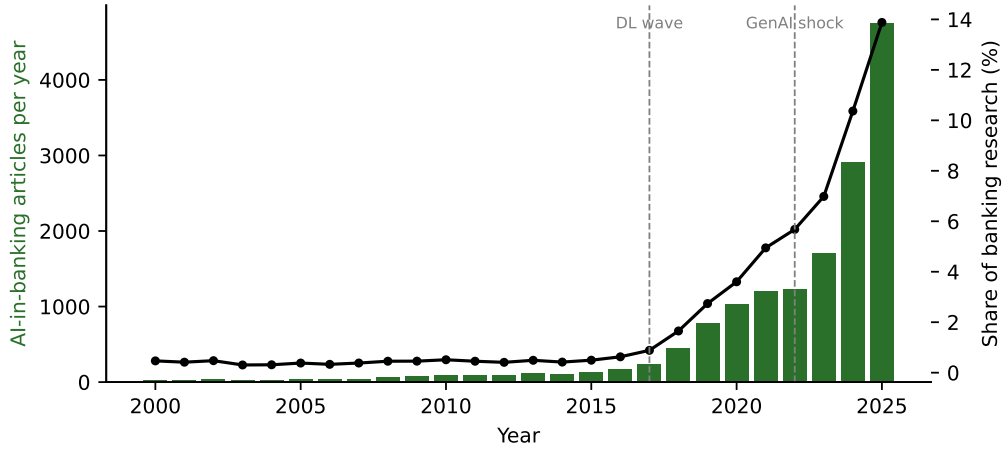


Figure 1: AI research in banking, 2000–2025. Bars: AI-relevant articles per year (left axis). Line: AI share of all banking-domain articles (right axis). Dashed lines mark candidate structural breaks at the deep-learning wave (2017) and generative-AI shock (2022).

4. Results

4.1. RQ1: The Rise

The rise is large on every measure (Fig. 1). Annual AI papers grew from 27 in 2000 to 85 in 2010 and 236 in 2017 - then 774 in 2019, 1,207 in 2021, 2,904 in 2024, and 4,750 in 2025, a 176-fold increase over the window. Because banking publishing itself grew (N_t rose from 5,774 to 34,224 per year), the field-share series is the cleaner gauge: AI's share of banking research climbed from 0.47% (2000) to 0.89% (2017) and then to 13.9% (2025) - roughly a thirty-fold relative increase. Era-level summaries appear in Table 1. Mann-Kendall tests reject trendlessness for both counts ($z = 6.79$, $p \leq 10^{-5}$; Sen's slope +41.9 papers/year) and shares ($z = 5.29$, $p < 10^{-5}$; slope +0.13 percentage points/year; both exploratory per Section 3.3). Sen's slope - the median of pairwise two-point slopes - falls well below the least-squares trend because the series is convex: on an exponentially growing series most early-era pairs yield small slopes, so the median understates the end-period rate; the log-scale analysis in Section 4.3 models the growth directly.

4.2. Growth Dynamics

The cumulative series (Fig. 2) makes the acceleration explicit: 463 AI papers appeared in 2000–2010 (42/year), 943 in 2011–2017 (135/year), and

Table 1: Era summary for AI-banking output. CAGR computed on endpoint years of each era.

Era	AI papers	Papers/year	Endpoint CAGR	Share at end
2000-2010	463	42.1	12%	0.51%
2011-2017	943	134.7	18%	0.89%
2018-2025	14,045	1755.6	40%	13.88%

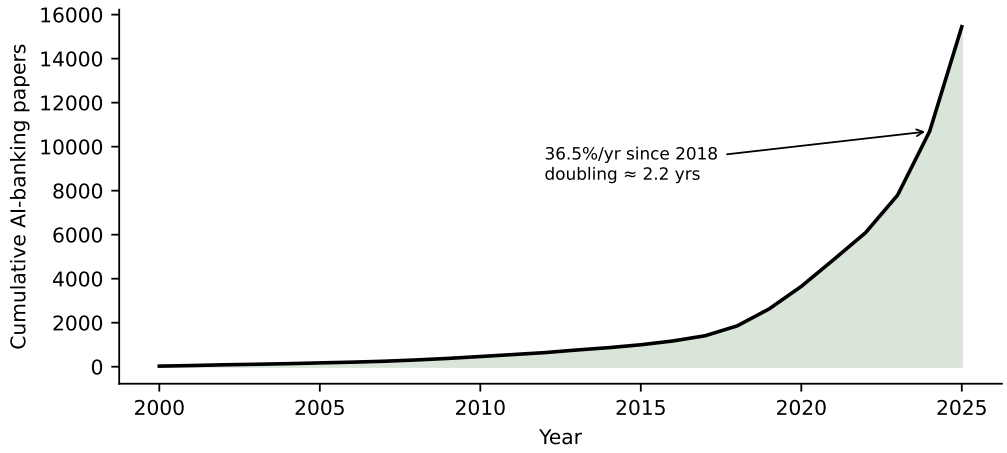


Figure 2: Cumulative AI-relevant banking papers, 2000–2025.

14,045 in 2018–2025 (1,756/year). On a log scale (Fig. 3) counts are approximately linear in time - exponential growth - with the slope steepening after 2022. In the modern era output compounds at 36.5% per year (2018–2024 CAGR), a doubling time of roughly 2.2 years. Endpoint growth rates by era rise from 12% (2000–2010) to 18% (2011–2017) to 40% (2018–2025); the modern-era figure is higher than the 36.5% above only because it uses the 2025 endpoint, the single largest year-on-year jump in the series, and should be read as an upper estimate of the era's typical rate.

4.3. RQ2: Structural Breaks

Growth was not smooth. Throughout, a break labelled T splits the series into a pre-break segment ending in T and a post-break segment beginning in $T + 1$, so the reported $F(2, 22)$ statistics use 24 residual degrees of freedom. Both pre-declared breakpoints are supported: Holm-corrected $p < 0.001$ on the asymptotic null, and $p \leq 0.0001$ on the wild bootstrap Table 3 that we

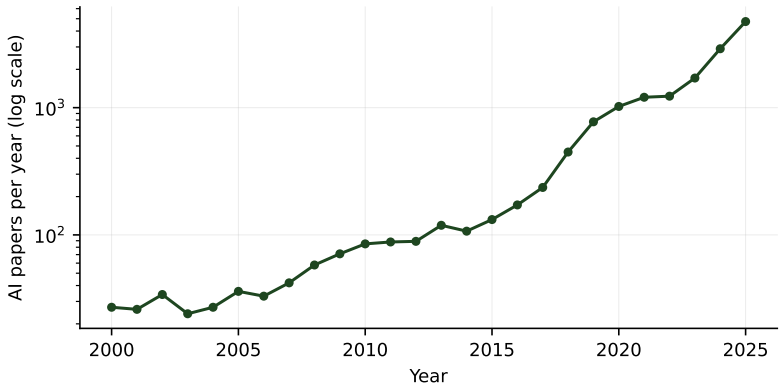


Figure 3: Yearly AI-banking output on a logarithmic scale. Linearity indicates exponential growth; the post-2022 slope increase marks the acceleration.

treat as decisive. For annual counts the 2017 break gives $F(2, 22) = 41.1$ and the 2022 break $F(2, 22) = 105.9$; the corresponding OLS segment slopes rise from +9.9 papers/year over 2000–2017 to +253 over 2018–2021 and +1,175 over 2022–2025. On field share the 2017 break gives $F(2, 22) = 133.9$ and the 2022 break $F(2, 22) = 48.0$, the latter visible as an acceleration from +1.07 percentage points/year (2018–2021) to +2.80 (2022–2025), a factor of 2.6.

A single-year Chow test does not identify a date here.. Both pre-declared tests reject, but rejection and identification are different things, and on this series the Chow statistic cannot supply a date by itself. Table 2 reports $F(2, 22)$ for every candidate split from 2013 to 2020 on all four specifications. Every one of the thirty-two tests rejects at conventional levels; the pre-declared 2017 date is not singled out. The reason is visible in the profile: on raw counts the statistic rises monotonically across the entire admissible range, from $F = 3.5$ at a 2005 split to $F = 105.9$ at 2022, placing its maximum at the boundary. That behaviour is what a convex series produces whatever the truth - put the split anywhere to the right of the curvature and the two fitted segments will differ more. Raw-scale F therefore largely measures where the series is most convex, not where a regime changed.

The curvature-free specifications behave differently and do identify a date. On log counts the statistic is flat from 2013 to 2016 ($F = 55.0, 54.5, 55.0, 57.5$), peaks at 2017 ($F = 61.5$) and then collapses ($F = 34.3$ at 2018, 15.9 at 2019); on log shares it peaks at 2016 ($F = 163.0$) having been essentially flat from

2013 ($F = 161.0$). Interior maxima of this kind cannot be manufactured by convexity. The two curvature-free specifications are also adjacent, and they bracket the pre-declared date. We therefore localise the deep-learning transition to the 2016–2017 window and claim a break interval rather than a break year. The pre-declaration remains worthwhile as a discipline - the date was fixed before estimation - but it should not be mistaken for identification, and the F profile would have supported a 2016 date just as well as a 2017 one.

Table 2: Chow $F(2, 22)$ at every candidate split, 2013–2020. The pre-declared date is marked. All tests reject at conventional levels; the informative content is the shape of the profile, not any single cell. The two log specifications have interior maxima (2017, 2016); the raw-count profile rises monotonically to the boundary and so cannot locate a date.

Series	2013	2014	2015	2016	2017	2018	2019	2020
Counts	19.9	24.5	29.8	35.6	41.1	47.2	55.8	70.1
Field share	43.4	59.9	82.7	110.2	133.9	147.9	137.6	107.1
Log counts	55.0	54.5	55.0	57.5	61.5	34.3	15.9	9.2
Log shares	161.0	161.1	162.3	163.0	138.9	58.0	25.5	14.2

Pre-declared date in bold. All 32 tests reject at conventional levels; the informative content is the shape of the profile, not any single cell. On raw counts the statistic increases monotonically across the full admissible range, placing its maximum at the boundary (2022, $F = 105.9$) - the signature of convexity rather than of a located break. The two log specifications have interior maxima.

Table 3: The confirmatory family: Chow statistics for the two pre-declared breakpoints, with asymptotic and wild-bootstrap p-values (20,000 draws, Rademacher weights, restricted residuals). The bootstrap values are the ones we rely on; it inflates the two attenuated 2022 log-scale tests without overturning them.

Series	Break	$F(2, 22)$	Asymptotic p	Wild p	Verdict
Counts	2017	41.1	$< 10^{-7}$	0.0001	reject
Counts	2022	105.9	$< 10^{-5}$	< 0.0001	reject
Field share	2017	133.9	$< 10^{-6}$	< 0.0001	reject
Field share	2022	48.0	$< 10^{-8}$	< 0.0001	reject
Log counts	2017	61.5	$< 10^{-7}$	< 0.0001	reject
Log counts	2022	5.5	0.011	0.020	reject
Log shares	2017	138.9	$< 10^{-7}$	< 0.0001	reject
Log shares	2022	5.2	0.015	0.017	reject

An endogenous scan probes whether the data would have found a break somewhere else. A supremum-F search over every admissible breakpoint 2005–2023 with wild bootstrap inference (1,000 replicates, so $p \leq 1/1001$ is the resolution floor) reproduces the profile in Table 2: the maximum sits at 2016 on log shares, 2017 on log counts, 2018 on raw shares, and at the 2022 boundary on raw counts. The two curvature-free maxima fall in the 2016–2017 window. Worth noting: 2020 ranks only third on raw counts and fifth on raw shares, so the pandemic-year discontinuity in banking publishing does not generate a break of its own. These scans are exploratory (Section 3.3); they corroborate the window identified above but cannot supply independent confirmation, because the statistic is the same one.

Functional-form sensitivity.. Because annual publication counts compound approximately exponentially (Fig. 3), we re-estimate every break test on logarithmic counts and log shares. The 2017 result is if anything sharper in this specification: the Chow statistic rises to $F(2, 22) = 61.5$ for log counts and 138.9 for log shares (wild-bootstrap $p = 0.001$ each), and the endogenous maximum for log counts sits exactly at 2017 ($\sup F = 61.5$). The 2022 result changes character. Its log-scale statistic is $F(2, 22) = 5.5$ on log counts (wild-bootstrap $p = 0.020$) and $F(2, 22) = 5.2$ on log shares ($p = 0.017$) - both still rejecting, but an order of magnitude smaller, and on the log profiles both sit on the far side of the interior maximum rather than near it. The bootstrap inflates these two p-values relative to the asymptotic null (0.011 and 0.015) without overturning them, which is the expected direction and the reason we report both. A Bai–Perron-style grid search over all breakpoint pairs confirms the asymmetry: for log shares the preferred pair is 2015/2018 ($F = 135.9$), while the pre-declared pair (2017, 2022) ranks only 66th of 120 admissible combinations - barely better than the median pair, so the rank itself carries little evidential weight - and for log counts the three highest-ranked pairs all place their second break at 2017, with 2022 absent from the top four. The endogenous log-share maximum likewise moves into the mid-2010s (2016; $\sup F = 163.0$). We therefore read the record as follows: a multiplicative-growth regime change located in the 2016–2017 window on the two curvature-free specifications, followed after 2022 by a significant but composition-centred acceleration whose evidence lives in levels, topic reallocation (Section 4.4), and venue dynamics rather than in a clean change of trend growth. The Bai–Perron grid is consistent with this reading without being decisive: for log shares the preferred pair is 2015/2018 ($F = 135.9$),

which brackets the same window from outside, while the pre-declared pair (2017, 2022) ranks only 66th of 120 admissible combinations - barely better than the median pair, so the rank carries little evidential weight on its own. For log counts the three highest-ranked pairs all place their second break at 2017, with 2022 absent from the top four.

Venue composition of the post-2022 surge.. Version duplication and low-barrier outlets could mechanically inflate post-2022 growth, so it matters whether the surviving signal is produced by them. It is not. Preprint servers account for 2.7% of 2022–2025 AI-classified output - down from 5.2% in 2017–2021 - so preprint-version duplication cannot account for the surge; rapid-review multidisciplinary outlets rise from 0.7% to 5.3%. The remaining 92% of output flows through established journals and other venues, and within that restricted stratum the log-scale 2022 statistic is essentially unchanged (Chow on log counts excluding preprints: $F(2, 22) = 6.0$, $p = 0.008$; excluding preprints and rapid-review outlets: $F(2, 22) = 5.6$, $p = 0.011$, against 5.5 on the full sample). The point of that comparison is persistence under exclusion, not magnitude: all three are the same attenuated signal, and none of them is produced by preprint or rapid-review venues. It does not speak to the stable-venue panel below, which is a different restriction and a different result.

Denominator sensitivity.. Field share is a ratio, so a change in the denominator moves it even when the numerator is unremarkable. Banking-domain output is not monotone: it peaks at 28,373 records in 2020, contracts to 24,403 in 2021 and 21,680 in 2022, and does not regain its 2020 level until 2025 (2024 sits 362 records below the peak). Holding the denominator at its 2020 value and recomputing the share series reduces the 2021-to-2022 share gain from +0.73 to +0.08 percentage points, while the 2024-to-2025 gain rises, since total banking-domain output expands 22.2% year on year. The consequence is specific: on the share measure, most of the apparent 2022 discontinuity coincides with a contraction in total banking publishing and is therefore not independent evidence of a research reallocation. The counts-based break is unaffected, which is why it carries the weight of the generative-AI claim. The 2020 base year is the observed peak and so is a post hoc choice; a symmetric alternative is to interpolate N_t linearly across 2020–2022, which yields the same qualitative picture, since AI counts rose in each of those years (1,022, 1,207, 1,231) while N_t fell.

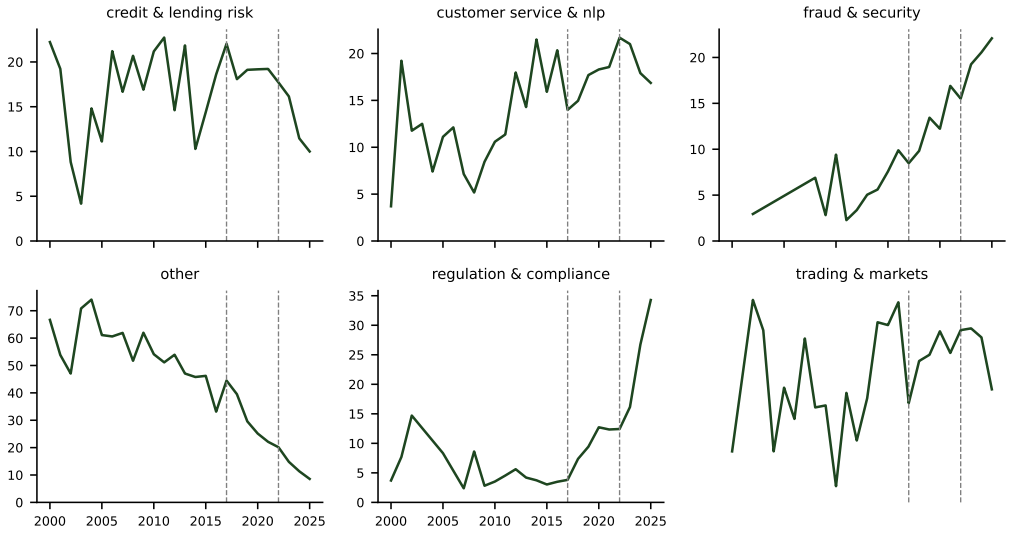


Figure 4: Topic-bucket shares over time (small multiples). Dashed lines: candidate breaks at 2017 and 2022.

4.4. RQ3: Topic Migration

Composition shifted decisively from general to applied (Fig. 5). Unclassifiable "other" work fell from 50.3% of AI output (2000–2016) to 28.1% (2017–2021) to 11.7% (2022–2025), while identifiable application areas grew: regulation and compliance rose six-fold (4.3% to 26.7%), fraud and security four-fold (5.0% to 20.4%), and customer service and NLP climbed steadily (14.2% to 18.4%). Credit and lending risk - long the largest named area - declined relatively (17.1% to 12.3%) even as it grew in absolute terms (from 197 to 1,303 papers). The post-2022 period therefore marks not only more AI research but a different kind: increasingly regulatory, security-oriented, and language-centric. Per-topic trend lines with both candidate breaks are shown in Fig. 4; testing each bucket's 2022 break individually ($F(2, 22)$, uncorrected across the six buckets) indicates that only regulation and compliance shows a structural change at that date ($F = 20.8$, $p < 0.001$), localizing the post-2022 shift in the literature to compliance research. The correction matters: with six buckets tested, this p would not survive Holm adjustment, so the localization is suggestive rather than established.

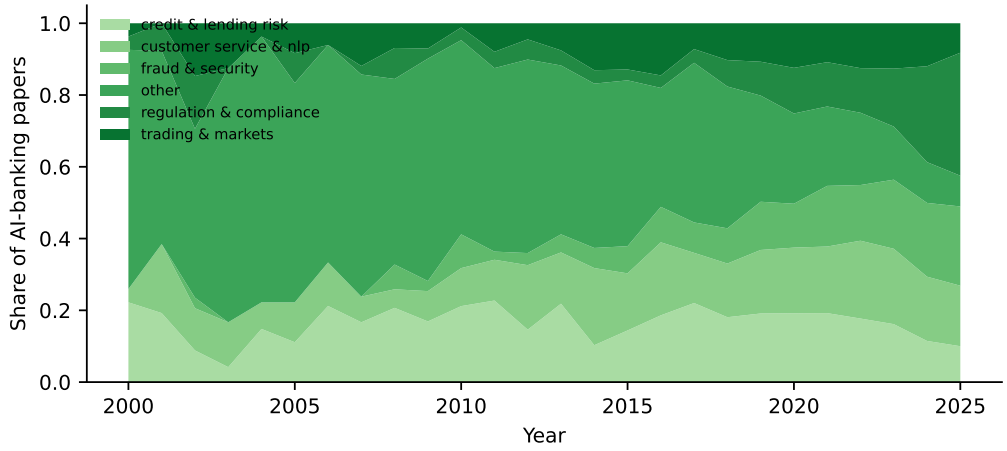


Figure 5: Topic-bucket shares of AI-relevant banking papers by year.

4.5. Robustness and Sensitivity

Four checks probe dependence on analytical choices. First, an expanded AI term list (adding reinforcement learning, random forest, support vector machines, gradient boosting, chatbot, robo-advisory, expert systems, text mining, computer vision, generative AI, and ChatGPT) raises the AI subset from 15,451 to 17,050 papers (+10%) while the yearly share series is nearly unchanged; Mann-Kendall and both Chow tests retain significance with nearly identical statistics (Table 4). Second, the per-bucket break tests of Section 4.4 localize the 2022 shift to compliance research, and that localization is the one per-bucket result that would not survive multiplicity correction across the six buckets - so we treat the compliance concentration as suggestive. Third, restricting the search to article titles rather than titles plus abstracts changes corpus size but not the ordering of yearly shares. Fourth, a coverage-drift probe addresses the concern that Semantic Scholar progressively ingested journals over time, so that measured growth could partly reflect database expansion. We construct a stable-venue panel of 38 outlets with at least 10 banking-domain papers in 2000–2009 *and* at least 25 in 2018–2025 (12,227 papers). Within this panel the 2017 regime change is fully preserved (log counts: $F = 13.7$, wild-bootstrap $p = 0.001$; log shares: $F = 8.0$, $p = 0.001$), while the 2022 acceleration disappears (log counts: $F = 0.0$; log shares: $F = 0.8$, $p = 0.46$).

The panel needs a caveat that cuts against our own use of it. Its second

criterion conditions on late-period volume, so it selects outlets that stayed busy through 2018–2025; a venue excluded for having contracted is absent by construction. The panel therefore does not by itself show that the 2017 growth is free of ingestion artifact, only that it is present among outlets that remained active. A pre-period-only criterion would be the cleaner test, and we flag it as the design change most worth making in a replication.

A window-truncation test addresses a different concern and is computable from the aggregate series alone: could the 2017 break be manufactured by the post-2022 surge pulling the fit? It is not. Re-estimating the 2017 break on log counts with the series truncated at each successive year end gives $F = 32.0$ (2000–2020), 40.0 (2000–2021), 40.1 (2000–2022), 46.0 (2000–2023) and 53.9 (2000–2024), all at $p < 0.001$ on their respective degrees of freedom. The statistic is stable across every window end and never approaches the conventional threshold, so the 2017 result does not depend on the presence of the 2022 surge in the data. The 2022 break, by contrast, cannot be estimated at all on windows ending before 2022, and on windows ending 2023–2024 it falls outside the admissible range; its estimate is therefore contingent on the 2024–2025 data, as discussed in the data-availability note.

Read alongside the preprint decomposition above, the two composition controls together localise the 2022 surge more precisely than either does alone. Preprints account for a *falling* share across the break (5.2% to 2.7%), and rapid-review outlets for only 5.3% of 2022–2025 output, so the surge is not a versioning or low-barrier-publication artifact. It also vanishes within the long-established panel. The implication is that the growth concentrates in mid-tier, long-existing, non-preprint outlets: journals with modest early-period volume that were nonetheless already indexed. This is a narrower and more mundane claim than the one we would need to support a compositional explanation of the kind usually offered for generative-AI-driven bursts, and we state it in preference to that claim. Taken with the denominator sensitivity in Section 4.3, the evidence for a discrete 2022 regime change is materially weaker than the 2017 result, and we report it as a real but modest acceleration rather than a confirmed break.

4.6. RQ4: Citations and Venues

AI papers earn a citation premium in every era examined. Median citations for era cohorts aged at least three years were 4 vs. 2 for non-AI peers in 2000–2010 (ratio 2.0), 4 vs. 1 in 2011–2017 (ratio 4.0), and 5 vs. 1 in 2018–2021 (ratio 5.0); every Mann-Whitney comparison rejects equality ($p \leq 10^{-5}$,

Table 4: Sensitivity of field-share trend and break statistics to the AI term list. Broadening the term list by 11 further families moves the subset size by 10% and the break statistics by under 2%.

Term list	Papers	MK z	Slope	2017	2022
Base (8 terms)	15,451	5.29	+0.13	133.9	48.0
Expanded (19 terms)	17,050	5.42	+0.15	136.7	48.6

z is the Mann-Kendall statistic on field share; slope is in percentage points per year. The last two columns report field-share Chow $F(2, 22)$ at each break; for the base list the log-scale equivalents are $F = 61.5$ (2017) and $F = 5.5$ (2022).

exploratory per Section 3.3). The AI medians are essentially flat across the three cohorts (4, 4, 5) while the non-AI medians fall toward the floor (2, 1, 1), so the widening ratio reflects a falling comparison baseline rather than a rising advantage. On medians of this size a single citation moves the ratio by half or more; the robust reading is "several times higher in every era," not a trend in the size of the advantage. Venue concentration moved in the opposite direction: the number of venues publishing AI-banking work expanded from 172 (2000–2010) and 243 (2011–2017) to 5,663 (2018–2025), while the Gini coefficient rose from 0.149 [95% CI: 0.092, 0.207] through 0.219 [0.112, 0.340] to 0.497 [0.461, 0.532]. Both movements are partly mechanical: a Gini over a growing set of units is diluted by the addition of many small venues, and the era windows differ in length. The concentration figures therefore describe the shape of the modern-era landscape rather than establishing a trend in concentration over time.

Citation distributions by era are shown in Fig. 6. The ten most prolific venues of the modern era (Table 5) are not traditional peer-reviewed banking journals. Of the 1,252 papers they contain, 469 (37.5%) sit in two preprint servers (arXiv, SSRN), a further 442 (35.3%) in four outlets characteristic of the rapid-review market, and 341 (27.2%) in the remaining four titles, one of which is IEEE Access - a fully indexed and widely read venue that we include in the residual rather than classify as low-barrier. Grouping the first two categories gives 72.8% of the top-ten concentrated output sitting outside conventional journal review, which is a composition effect readers of the AI-banking literature should weigh when sampling it. This concerns the outlets that dominate the ranking, not the field as a whole: the 5,663 venues of the modern era collectively include the established banking and finance journals, and this table says nothing about their quality. The Lorenz curves

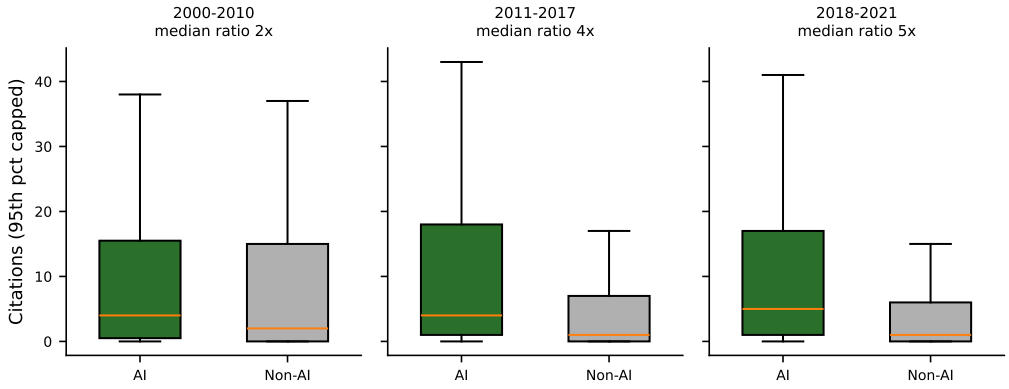


Figure 6: Citation distributions for AI vs. non-AI banking papers by era cohorts aged ≥ 3 years (whiskers omit outliers above the 95th percentile).

behind Fig. 7 show the same asymmetry continuously rather than at the top of the ranking.

Because AI and non-AI papers publish in systematically different outlets, the pooled premiums above could partly reflect venue prestige rather than topic. A within-venue reanalysis (venues with ≥ 20 AI and ≥ 50 non-AI papers in a cohort; 10 venues qualify for 2018–2021, one for 2011–2017, none earlier) shows the premium survives matching but attenuates: median within-venue ratio of 1.7 for 2018–2021, positive in 8 of 10 venues and individually significant in 6, against a pooled ratio of 5.0. The direction of travel is clear, but the two figures are not commensurable: one is a median of per-venue ratios, the other a ratio of pooled medians, so their difference does not measure the share of the premium attributable to venue. With one qualifying venue before 2018, we treat this as evidence that the premium is not purely a prestige artifact rather than as a decomposition of it.

5. Discussion

5.1. Interpreting the Breaks

Both detected breaks coincide with supply-side technology shocks outside banking, which is the pattern our design is built to detect; we have no demand-side control, so we cannot exclude a coincident banking-specific event. Post-2017 growth coincides with the mainstreaming of deep learning and transformer architectures [37, 18], which lowered the cost of building useful

Table 5: Ten most prolific venues for AI-banking papers, 2018–2025. Semantic Scholar records carry several source misspellings ("INTERANTIONAL" for *International*, inconsistent capitalisation) and long names truncated mid-word in the database; venue names here are lightly normalised and abbreviated for column width. The underlying strings are preserved in the deposited data.

#	Venue	Papers
1	arXiv.org	307
2	Social Science Research Network	162
3	World Journal of Advanced Research and Reviews	124
4	IEEE Access	120
5	Int. J. Research Applied Science & Engineering	108
6	Int. J. Multidisciplinary Research	106
7	Int. J. Scientific Research in Engineering	92
8	Int. J. Science & Research Archive	81
9	J. Information Systems Engineering & Management	79
10	Int. J. Advanced Research in Science & Comm.	73

financial models. Post-2022 acceleration coincides with public availability of instruction-tuned large language models [29] - reflected in our data by the rise of the customer-service/NLP and compliance buckets, the two areas where language technology bites first. The two transitions nonetheless differ in kind, and in two respects the paper's own data qualify the straightforward reading.

The first concerns the pre-2022 transition. We locate it in the 2016–2017 window rather than at a single year, because the Chow statistic cannot support a finer claim: on a convex series it rejects at every candidate date, and it is only on the curvature-free log specifications that an interior maximum appears (Section 4.3). That the interior maxima fall at 2016 and 2017 - adjacent years, straddling the mainstreaming of deep learning and transformer architectures - is the evidential content, and it is weaker than a headline break date but harder to dismiss. The transition is stable when the series is truncated at every year end from 2020 to 2024, so it does not depend on the post-2022 surge being present in the data (Section 4.5).

The second concerns the post-2022 signal, which is better described as a level-and-composition shift: its log-scale trend statistic is an order of magnitude smaller ($F = 5.5$ on log counts, 5.2 on log shares), and pairwise break searches prefer mid-2010s solutions - yet the acceleration itself persists in levels and survives exclusion of preprint and rapid-review venues ($F = 5.6$, essentially unchanged from 5.5 on the full sample, which is the point: the acceleration is not an artifact of those outlets) and localizes in compliance research (Section 4.4). It does not, however, survive restriction to long-

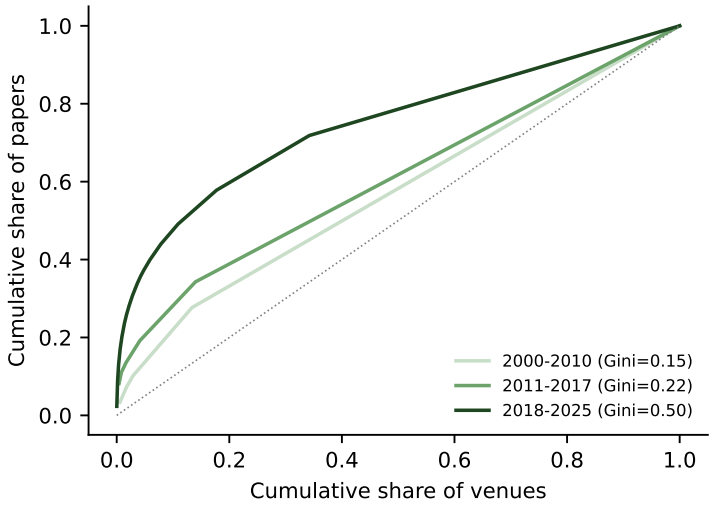


Figure 7: Lorenz curves for venue concentration of AI-banking papers by era.

established journals, where the 2022 acceleration disappears entirely (log counts $F = 0.0$; log shares $F = 0.8$, $p = 0.46$; Section 4.5). Unlike the 2017 break, the 2022 result cannot be estimated on windows truncated before 2022 and is sensitive to the 2024–2025 data. Taken with the denominator sensitivity below, the honest summary is that post-2022 AI publishing rose measurably, but the claim that it constitutes a distinct statistical regime is supported by the counts series alone and is not corroborated by the venue-composition evidence.

A caveat qualifies the 2022 result on the share measure. Banking-domain output fell by 23.6% between 2020 and 2022, from 28,373 to 21,680 records, so part of the post-2022 jump in field share is arithmetic rather than substance (Section 4.3). This is why we treat the counts-based break, which is immune to the denominator, as the primary evidence for a generative-AI acceleration and read the share break as corroborating rather than independent support. Generative AI appears to have acted first through *what* banking researchers study, rather than through an abrupt change in how much they publish.

The 2017 result, read against the canonical signature of a general-purpose technology diffusing into an applied field [9], matches the middle stage of that pattern: slow accumulation while core methods mature, then a sharp adoption break once the method base crosses a usefulness threshold. The later

stages are where our evidence thins. The literature anticipates composition effects concentrating where a technology's capability profile intersects domain demand; if that is what post-2022 banking research looks like, we would expect the growth to arrive through newly created outlets and through the sudden appearance of previously absent topics. Our venue evidence points elsewhere - to mid-tier, long-existing, non-preprint outlets (Section 4.5) - and our per-bucket evidence is suggestive rather than established once multiplicity is accounted for. We therefore describe the composition shift as real in direction and unestablished in mechanism. The finding that exposure to language models concentrates in analytical, higher-skill occupations [16] suggests that the supply of practitioner-researchers able to write about these tools may continue to grow; on that reading, the compliance-localized shift we measure would be a leading edge of a realignment of the field's attention rather than a passing response to a novel artifact. We note that this is a projection, not a finding: our data end in 2025 and contain no information on what follows.

5.2. Maturation, Premium, and Democratization

Three secondary patterns cohere. First, the collapse of unclassifiable "other" work from 50.3% to 11.7% of output signals maturation: early AI-banking research was generic method-transfer; current research addresses named problems. Second, the widening citation premium ($2\times \rightarrow 5\times$) should not be read as a rising advantage. The AI medians are essentially flat across the three cohorts (4, 4, 5); the ratio widens because the non-AI median falls toward the floor of the distribution (2, 1, 1). The defensible claim is that AI papers attract citations at a higher rate than their non-AI peers in every era examined, not that their advantage has grown over time. Third, rising venue count alongside rising cross-venue concentration (Gini $0.15 \rightarrow 0.50$) depicts a field spreading to thousands of outlets while its largest venues absorb a growing share of output. Note that both movements are partly mechanical in the unit count: more venues mechanically dilutes any Gini coefficient, and the era windows differ in length, so the figure supports a descriptive statement about the modern-era landscape rather than a precise concentration trend. The citation result deserves the same caution from the opposite direction. Work comparing national science systems finds that expansion of publication volume does not reliably coincide with rising relative citation impact [2]; our finding that AI papers are cited at several times the rate of their non-AI peers, without a corresponding rise in their absolute median, is consistent with that pattern rather than anomalous within it.

5.3. Practical Implications

For researchers: regulation and compliance is the fastest-growing application frontier, suggesting room for technical contributions where legal reasoning meets machine learning. For editors and reviewers: published volume in this space has been compounding for the better part of a decade, so evaluation criteria grounded in the pre-2022 literature may misjudge current norms. For policymakers: AI-classified banking papers are cited at several times the rate of their non-AI peers within the same era and often the same outlet, which is a reason to track quality control as volume grows rather than an argument about attention shares.

5.4. A Future Research Agenda

Six extensions follow directly from the data. First, the stable-venue panel of Section 4.5 should be rebuilt on a pre-period-only criterion - all outlets with at least ten banking-domain papers in 2000–2009, with no threshold on late-period volume - and accompanied by a decomposition of the 2022 surge by each venue's early-period volume band. Our current design cannot distinguish growth concentrated in mid-tier long-standing journals from growth concentrated in outlets that were already large, and that distinction determines how much of the 2022 result is a composition effect. Second, the compliance-localized shift (Section 4.4) invites fine-grained study of what LLM-assisted compliance research actually does: our keyword buckets cannot separate genuine methodological innovation from re-labeled prior work; full-text reading of a stratified sample of post-2022 compliance papers could. Third, the citation comparison would be considerably stronger with a floor-robust statistic: the share of papers with at least one citation, or Cliff's δ, computed within era cohorts. Our medians of 1–5 make the reported ratios fragile, and the widening we describe is a baseline effect rather than a growth in advantage (Section 4.6). Fourth, the venue-quality mix calls for studies linking outlet type to downstream citation quality and replication, testing whether the concentration of modern output in preprints and rapid-review outlets changes evidence standards in banking research - the venue-matched analysis in Section 4.6 is a first step that should be extended with venue fixed effects on article-level outcomes. Fifth, extending the series forward and across adjacent domains (insurance, fintech broadly) would establish whether the 2022 break is a banking-specific response or the research-system signature of a general-purpose technology; cross-sector comparisons at matched event dates are the natural design. Sixth, the classification layer itself can be

upgraded: an embedding-based or LLM-labeled classifier, validated against our audit sample, would tighten precision while preserving comparability if re-scored against the same frozen corpus. The structural-break framework used here also applies unchanged to other method-driven shifts - blockchain after 2015, machine-learning risk models after 2010 - and replication on those series would show whether research systems respond to capability waves with a common timing signature.

5.5. Threats to Validity

Construct. Keyword classification misses unlisted terminology and admits marginal papers. A stratified audit of 100 randomly sampled AI-classified records (30 pre-2017, 30 for 2017–2021, 40 post-2022; sample and labels in the audit materials), labeled against a strict dual criterion - an AI method central to the contribution *and* banking/finance as the central domain - found pooled precision of 55% (95% Wilson CI 45–64%): 60% (42–75%) pre-2017, 43% (27–61%) in 2017–2021, and 60% (45–74%) post-2022. The era gradient is not statistically significant ($\chi^2(2) = 2.36$, $p = 0.31$), so the temporal-invariance assumption behind treating misclassification as symmetric across years is not rejected; the mid-period dip traces mainly to computer-science proceedings entries whose metadata mentions finance incidentally. Absolute AI counts are therefore upper bounds on true volume - roughly 8,500 rather than 15,451 papers at point estimates. Direction is more robust than level, but the distinction is worth stating precisely: a non-differential error that scales each year's numerator by a similar factor leaves break dates and composition shares approximately intact while leaving the magnitude of the trend biased. The audit supports the claim that the era gradient in precision is not detectable, not the stronger claim that classification is unbiased. Stricter screening (human-in-the-loop or embedding-based filtering) is the clearest route to tighter estimates. **Internal.** Citation premiums compare cohorts at different ages only up to 2021 cohorts; within-era comparison avoids cross-era age confounds but not field-size effects on citing behavior. The within-venue decomposition rests on ten venues in 2018–2021 and one earlier, which is too thin to support a quantitative split of the premium. **External.** One database (Semantic Scholar), one domain (banking), English-language works; findings may not transfer to other fields or languages. Field share is also sensitive to its denominator, which contracted sharply in 2021–2022 (Section 4.3); we treat the counts series as primary throughout for that reason. **Statistical.** Breakpoints were chosen ex ante from technology history, so Chow tests here

confirm timing rather than discover it; other candidate dates are untested by design. Annual series of this length exhibit positive serial correlation, which inflates parametric significance for trend and break statistics; wild bootstrap inference mitigates this for the headline tests, but corrected Mann-Kendall variants [19] and HAC covariance estimators would tighten the remaining estimates. With 26 annual observations, break tests also carry limited power against gradual curvature - one reason the log-scale re-estimation changes the picture for 2022.

A note on what our design can and cannot distinguish. Structural-break detection on bibliometric series is, to our knowledge, novel in this literature, and that novelty cuts both ways: the machinery is standard in macroeconomics [31, 38, 8], on inference about unknown break dates [3, 20], and in work on asymptotic growth in science systems [36], but it has not been applied to establish whether growth in a research field contains statistically detectable discontinuities. The contribution is therefore a method transfer, and the claims that follow from it are about the timing of change in publication output. They are not claims about the intellectual content of that output, about research quality, or about how much AI is actually used in banks - questions that require full-text reading and outlet-level evidence rather than counts. Where the two readings diverge, as they do for the pre-2017 period, we prefer the conservative one.

6. Conclusion

Drawing on 498,958 banking-domain articles (2000–2025), this study measured the rise of AI research in banking rather than asserting it. AI output grew 176-fold in count and from 0.5% to nearly 14% of field share. Two findings qualify how far the underlying statistic can be pushed, and both cut against the paper's original framing rather than for it. First, a Chow test on a convex series rejects at essentially any split: every candidate date from 2013 to 2020 rejects on all four specifications, and on raw counts the statistic rises monotonically across the whole admissible range. What identifies the deep-learning transition is not the raw-scale maximum but the interior maximum that appears once curvature is removed, at 2016 on log shares and 2017 on log counts. We accordingly report a transition window rather than a break year. Second, the post-2022 generative-AI acceleration is weaker than its $F(2, 22) = 105.9$ suggests: the statistic falls by an order of

magnitude on logs, disappears within the stable-venue panel, and coincides with a 23.6% contraction in total banking publishing.

What survives those qualifications is the substantive claim. A growth-regime change in AI-banking research dates to roughly 2016–2017, is stable across every truncation and specification we tried, and coincides with the mainstreaming of deep learning. Post-2022 publishing rose substantially, but through a composition shift - from generic method transfer toward regulation, security, and language applications - whose exact boundary the data do not settle. AI papers are cited at several times the rate of their non-AI peers within every era, though the AI median itself is flat across cohorts, so this is a level difference rather than a widening advantage. The venue landscape broadened even as its leadership concentrated. Every figure regenerates from cached data through a version-controlled analysis pipeline.

Two caveats bound these conclusions. Pooled classification precision is 55% (95% Wilson CI 45–64%, $n = 100$) and a single bibliographic database cannot capture all of scholarly publishing. Within those bounds the direction of travel is clear, and its steepest segment is the most recent - so any snapshot predating 2022 systematically understates where the field now stands. Future work should replicate the break analysis on curated databases, extend classification beyond keywords, and couple publication trends to deployment evidence inside financial institutions.

Appendix A. Yearly Counts

Table A.6 reports the full yearly series underlying all analyses.

Appendix B. AI-Detection Terms

The eight term families defining the numerator - the set of records counted as AI-relevant - are frozen before analysis and applied identically to all years. Matching is case-insensitive substring containment against the concatenated title and abstract; the "(s)" notations in Table B.7 are literal and so match both the singular and plural forms. A record matching any family enters the numerator once.

Appendix C. Analysis Windows

Several distinct partitions of the 26-year series appear in this paper, and using the word "era" for all of them would obscure which is which. The

Table A.6: Yearly banking-domain output (N_t), AI-relevant output (A_t), and AI share, 2000–2025.

Year	N_t	A_t	Share	Year	N_t	A_t	Share
2000	5,774	27	0.47	2013	24,242	119	0.49
2001	6,249	26	0.42	2014	25,460	107	0.42
2002	7,145	34	0.48	2015	26,673	132	0.49
2003	7,852	24	0.31	2016	27,344	172	0.63
2004	8,629	27	0.31	2017	26,655	236	0.89
2005	9,434	36	0.38	2018	27,104	448	1.65
2006	9,981	33	0.33	2019	28,255	774	2.74
2007	10,983	42	0.38	2020	28,373	1,022	3.60
2008	12,768	58	0.45	2021	24,403	1,207	4.95
2009	15,541	71	0.46	2022	21,680	1,231	5.68
2010	16,603	85	0.51	2023	24,470	1,709	6.98
2011	19,393	88	0.45	2024	28,011	2,904	10.37
2012	21,712	89	0.41	2025	34,224	4,750	13.88

following names are used consistently:

- **Eras** (growth, venues, concentration): 2000–2010, 2011–2017, 2018–2025; see Table 1.
- **Topic windows** (the six application buckets): 2000–2016, 2017–2021, 2022–2025, set so the middle window ends before the 2017 break and the last begins at it.
- **Audit strata** (precision audit): pre-2017, 2017–2021, post-2022.
- **Citation cohorts** (truncated to limit right-censoring): 2000–2010, 2011–2017, 2018–2021.

Appendix D. Application-Topic Bucket Definitions

Application-topic buckets are assigned by longest-keyword-match against the lowercased concatenation of the record's primary-topic string, title, and abstract; records matching no keyword are labeled *other*. Table D.8 discloses the full keyword lists. Assignment is deliberately transparent rather than semantic, and is applied identically across years. Two consequences are worth stating. Because the rule is longest-match over a shared text, a paper naming

Table B.7: The eight frozen AI-detection term families defining the numerator. These are the strings a reader needs in order to reproduce the classification exactly.

#	Term family
1	artificial intelligence
2	machine learning
3	deep learning
4	neural network(s)
5	large language model(s)
6	natural language processing
7	data mining
8	predictive model(s)

both a market and a compliance term lands in whichever keyword string is longer, so bucket boundaries are approximate. And because the keywords are matched as substrings, "regulat" also fires on *regulatory* and *regulation* but not on the noun *rule*, while "text" fires on *context* - which is why the customer-service bucket should be read as a proxy for language-oriented work rather than a clean content label.

Table D.8: Keyword families defining each application-topic bucket (longest match wins; *other* if no match). Matching is case-insensitive substring containment, so short keywords can capture unintended words (*text* within *context*).

Bucket	Keywords
Credit & lending risk	credit, loan, default, lending, scoring
Fraud & security	fraud, anomaly, cyber, security, aml, money laundering
Trading & markets	stock, trading, portfolio, market, price, forecast
Customer service, NLP	chatbot, sentiment, text, nlp, language model, customer
Regulation & compliance	regulat* (stem), compliance, supervision, risk management, governance
Other	(no keyword matched)

Trading and markets is an explicit bucket, reported here as a residual share: 9.1% in 2000–2016 and 10.5% in 2022–2025.

Author Contributions

Md. Mahin Ahmed: Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Institutional Review Board Statement

Not applicable. The study is a bibliometric analysis of published bibliographic records and involves no human participants, no intervention, and no collection of personal or sensitive data. Ethical approval was therefore neither sought nor required under the institutional policies governing secondary analysis of publicly available scholarly metadata.

Informed Consent Statement

Not applicable, for the reasons given above.

Acknowledgments

The author thanks the maintainers of the Semantic Scholar open data platform [25] and of Bibliometrix [4], on whose conventions and tooling parts of this analysis depend.

Conflicts of Interest

The author declares no competing financial or personal interests that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the author used an AI coding agent (Claude, used programmatically through a command-line interface) to assist with routine elements of the analysis workflow, including code drafting, cross-checking of reported statistics, and reference verification. The study design, the choice of measures and tests, the interpretation of results, and all conclusions are the author's own. All analyses were executed against the frozen corpus, every number in the manuscript was recomputed from that corpus during revision, and the author reviewed and verified the manuscript in full, taking complete responsibility for its content.

Data availability

The analysis corpus (identifiers, titles, abstracts, venues, citation counts), the frozen search queries, all pipeline code with unit tests, the manual-audit sample, and every result file underlying the tables and figures are available from the author upon request and will be deposited in a public repository with a permanent DOI upon acceptance.

Corpus provenance

The denominator corpus is defined by the frozen query given in Section 3.1, restricted to works published 2000–2025 and retrieved with an exhaustive pagination fan-out against the Semantic Scholar bulk-search API. The response set was written to disk once, with retrieval dates recorded, and never re-queried; all tables and figures regenerate from that cache. One design limitation deserves statement: a corpus of this size cannot be assembled from the public unauthenticated endpoint without such a fan-out, so coverage is a property of the fan-out schedule as much as of the database. Per-year record counts in Table A.6 are sufficient to detect coverage drift between replications.

Right-edge truncation

The final year carries the headline 13.9% share and the 4,750 count, and indexing lag makes the most recent year the least complete. Re-estimating both pre-declared breaks through 2024 only, dropping that year, leaves the 2017 break comfortably supported (on counts $F(2, 21) = 71.9$, on field share $F(2, 21) = 217.0$, and on the log transformations $F(2, 21) = 53.9$ and 116.4)

and leaves the 2022 break untouched, since it is interior to the series. Growth and concentration conclusions therefore do not depend on the completeness of the last observation; only the terminal share and the 176-fold multiple do.

Implementation and reproducibility

Analyses were performed in Python 3.14 with NumPy and SciPy. Random seeds for the wild bootstrap (1,000 replicates) and the Gini percentile bootstrap (2,000 resamples, resampling *venues*, since venues are the Gini's units) are fixed in the released configuration; the wild bootstrap generates its null from restricted-residual distortions under the null of no break. Bootstrap p-values from 1,000 replicates are reported as $p \leq 1/1001$, their resolution floor, rather than as point estimates. We also note that the precision audit was carried out by a single rater without blinding and is therefore exposed to confirmation bias; second-rater agreement on a subset of the 100 records would strengthen it.

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